425282 Deep Learning

Final Project

Last update: 26 mar

Important Dates

Deliverable	Weight 50%	Due Date
Topic proposal	0	10th April (maybe sooner)
Milestone 1	5	24th April
Milestone 2	15	22th May
Poster session	15	3th june
Report	15	

Overview

One of the main goals of this course is to be able to apply deep learning methods to solve real problems with real data, or to be able to start deep learning research. In this project, you will apply the knowledge acquired during the semester to a problem of your interest. The topic is completely open, so feel free to choose one related to your background, your personal interests, your specific MSc field, or you may even start thinking of a topic for next year's dissertation. You can use any kind of data you want: images, videos, text, tabular data, time series, and others. It can be generative or discriminative.

Keep in mind that it should be feasible using the free tier of Google Colab, Kaggle computing, or similar infrastructures. Room 1.3.30 is equipped with GPUs RTX 3060 which you can use whenever the room is free - just ask the security downstairs, near Staples store, to open the door. We might be able to arrange for GPUs for a very limited number of experiences for each group.

This document will be updated continously with more information, sources, and guidelines.

Topic

Here are some sources you may want to consult to inspire your work:

- Last year projects: topics , posters
- [Computer Vision] Stanford's CS231n editions of 2024, 2022, 2017, 2016, 2015.
- Stanford's CS229 editions of 2022, 2021, 2020, 2019.
- Kaggle challenges
- Medical Imaging
    - Grand Challenge platform
    - https://github.com/beamandrew/medical-data
    - https://conferences.miccai.org/2023/papers/ or other miccai editions.
- Benchmark collection
- Awesome Colab Notebooks
- Challenge workshops of well-known conferences such as:
    - CVPR: workshops
    - ECCV, ICCV
    - Neurips and Neurips Datasets Track 2024 2023 2022, competitions
    - MICCAI,
    - KDD,
    - and so on.
- Comudel.com https://docs.google.com/document/d/1C9kColfy57M_6qRpLZnOXIZcgUCaD9t0_tClaUoU0QI/edit?usp=sharing
- You can email me asking for directions if you want. I will add concrete suggestions here.

Project stages

The very first task is to identify what you like and try to find a group with similar interests. You can use the dedicated forum on moodle, you can use the TPs to pitch your idea, the group self-selection activity on moodle, etc.

Topic Proposal

Your first task is to choose a topic and write a short document describing the fundamental aspects of the project. The goal is to receive feedback very early, so try to be as specific as you can at this point.
The document should be one page, including the following information:

- Title, group members
- What problem will be investigated, and why is it interesting?

- What sources will be reviewed to provide context and background?
- What data will be used? If new data is collected, how will it be gathered?
- What method or algorithm will be used? If existing implementations are available, will they be utilized, and how? While an exact answer is not required at this stage, a general approach to the problem should be considered.
- How will the results be evaluated? Qualitatively, what kinds of results are expected (e.g., plots or figures)? Quantitatively, what kind of analysis will be used to evaluate and/or compare the results (e.g., performance metrics or statistical tests)?

Milestone 1

Please add max 2 pages to the report of Milestone 0 with:

- data pipeline, including preprocessing
- baseline description: model name / architecture depth; input shape; loss; optimiser; training budget (epochs, LR, batch, GPU time).
- preliminary results and training curves
- known issues and next steps, including a detailed plan for the rest of the experiments

By the first milestone, it is expected that you have the data downloaded and ready to be used, the models clearly identified and a skeleton code to start working on.
This time, the document should be 3 pages max and it can include whatever is still relevant from the proposal. It should have:

- Title, group members
- Introduction: this section introduces your problem, and the overall plan for approaching your problem
- Problem statement: Describe your problem precisely specifying the dataset to be used, expected results and evaluation
- Technical Approach: Describe the methods you intend to apply to solve the given problem
- Figure describing the approach / methodology. Get inspiration from the papers that you're reading for your project. Some examples:
    - figure 1 of this paper or similar
    - check the posters here or go check LASIGE posters in C6 floor 2.
- Describe the experiments you are planning to do and preliminary results if you have them.

Please use this template: https://github.com/cvpr-org/author-kit/releases

Milestone 2

Please add max 2 pages to the report of the last Milestone with:

- training methods

- experiments
- results, including learning curves.
- contribution of each member of the group

Room 1.3.30 is operational again.

Poster

- The poster session will be held in the atrium of the C6 building starting at 16:30 on Wednesday, May 28th (same as lecture time).
- The poster should be size A1 portrait-oriented, coloured.
- You can use this template and change as you'd like. Just keep the acknowledgments and logo.
    - link
- You can print it on your own, or you can print it for free at Planeta Colorido - https://maps.app.goo.gl/FJ5VWEgk7aTBccPy7 .
    - just let them know you're coming from the Deep Learning class at FCUL.

What's a poster session: https://www.youtube.com/watch?v=oUqnvQm_k9M
The idea is that one member of the group walks around the poster session and gets to see what the other groups did, while the other stands next to the poster and continuously presents the work to those passing by. After some time you swap roles.
I'll be there making questions about your work as well. There might be people from companies attending the poster session, as well as colleagues from research labs and the department.
To get inspired for your poster, you can check posters here, or LASIGE Workshop - from here. -> posters or go check LASIGE posters in C6 floor 2.

Final report

- Please update the report submitted in the previous milestone to be consistent with the work actually done. If what is described in the milestone report is accurate, then you only need to describe the experiments and report the results. You can use the tables from the poster. If the methodology or datasets changed, please change the report accordingly.
- The report should be 8 pages max.
- Use this template: https://github.com/cvpr-org/author-kit/releases

Submission. Each group should submit:

- Report document (paper format; ideally, it should be what was submitted in the last Milestone + results tables from the poster. So, the amount of writing and effort to produce this document should be reduced, as almost everything is already written.)
- Poster (PDF)
- Code or link to code

Weight on final grade, as agreed in the beginning of the semester:

- 15% - corresponds to the report submitted on Moodle. There is a soft deadline (4 June) and a strict deadline on Friday (6 June).

Optional: Writing hackathon - Taking a project to publication

writing get together (class of 4th June)

- The goal is to finish the report during class. This is a "get together and write" session, where you can focus on writing and discuss with colleagues how best to present your results (e.g., Figure 1, abstract, result formatting).
- No coding is involved.
- Attendance is not mandatory. Still, the more groups present, the more productive the session can be in terms of exchanging ideas and refining presentations.
- Reference/inspiration: https://sph.umich.edu/cehr/pdf/Paper_Sprint_Manual.pdf

Maybe some of you (I hope many) would like to continue work in research. Even if that's not the case, maybe you'd like to experiment what it's like to go to a conference and present your work. It's also a good line in your CV, whether you plan to go to industry or to apply for a scholarship during the 2nd year of your masters.

The National Symposium of Informatics or RECPAD are great places to start, and the deadlines coincide with the end of the semester.
Let me know if you'd like to pursue this. The class of 4th June is also a good occasion to write the paper.

Useful link: Taking a Course Project to Publication - https://cs231n.github.io/choose-project/

Submission

Please submit a zip with the code, the report, and the poster file.

Groups

Please use the Moodle tool to form groups of 3 max. Groups of 1 are not allowed.